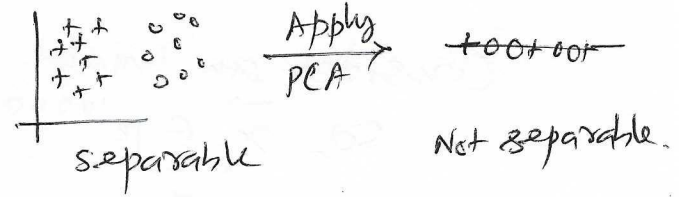
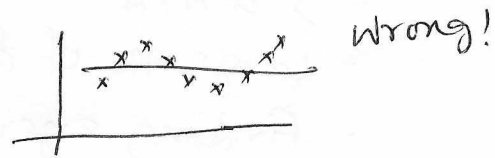


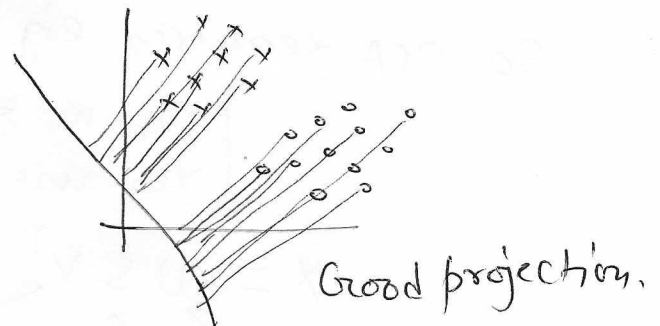
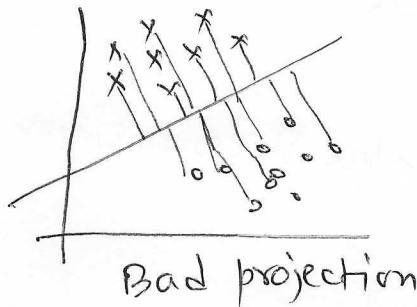
Linear Discriminant Analysis (LDA)

Problems with PCA

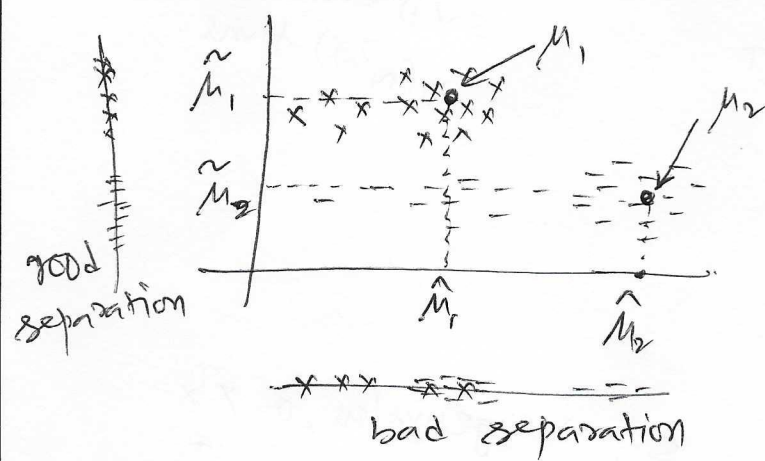
- Captures linear dependence only
- Direction of maximum variance may be useless for classification



Finding a good projection $\rightarrow$ LDA



How to find such a line for projection?



Large $|\tilde{\mu}_1 - \tilde{\mu}_2| \rightarrow$ better separation!
Not really, see the example.

problem

variance of the classes are not taken into consideration.

$|\tilde{\mu}_1 - \tilde{\mu}_2|$ should be normalized by the variance of the classes.

Let we have two classes C_1, C_2 having d -dimensional data $x_1, \dots, x_n$.
Where n_1 samples are from C_1 , & n_2 samples from C_2 .

For a data point x_i , its projection on the line having direction given by unit vector v is, $v^T x_i$ (scalar projection)



(33)

Mean of class $C_1 \rightarrow \mu_1$
 " " " $C_2 \rightarrow \mu_2$

$$\tilde{\mu}_1 = \frac{1}{n} \sum_{x_i \in C_1} v^T x_i$$

Projected " " " $C_1 \rightarrow \tilde{\mu}_1$
 " " " " $C_2 \rightarrow \tilde{\mu}_2$

$$= v^T \left(\frac{1}{n} \sum x_i \right) \\ = v^T \mu_1$$

$$\tilde{\mu}_1 = v^T \mu_1$$

$$\tilde{\mu}_2 = v^T \mu_2$$

Let $y_i = v^T x_i$ be the projected sample.Variance for samples of class C_1 & C_2 :

$$\tilde{s}_1^2 = \frac{1}{n} \sum_{y_i \in C_1} (y_i - \tilde{\mu}_1)^2$$

$$\tilde{s}_2^2 = \frac{1}{n} \sum_{y_i \in C_2} (y_i - \tilde{\mu}_2)^2$$

We need to project our data onto a line having direction v which maximizes:

$$J(v) = \frac{(\tilde{\mu}_1 - \tilde{\mu}_2)^2}{\tilde{s}_1^2 + \tilde{s}_2^2}$$

We need to write $J(v)$ in terms of v .Let the class covariance matrices for C_1 & C_2 are S_1 & S_2 :

$$S_1 = \sum_{x_i \in C_1} (x_i - \mu_1)(x_i - \mu_1)^T$$

$$S_2 = \sum_{x_i \in C_2} (x_i - \mu_2)(x_i - \mu_2)^T$$

$$\text{Now we can write. } \left. \begin{aligned} \tilde{s}_1^2 &= v^T S_1 v \\ \& \tilde{s}_2^2 &= v^T S_2 v \end{aligned} \right\} \begin{aligned} \tilde{s}_1^2 + \tilde{s}_2^2 &= v^T (S_1 + S_2) v \\ &= v^T S_w v \end{aligned}$$

↑
 how can we get
 this? Derive
 using projection
 geometry.

$$\text{Also, } (\tilde{\mu}_1 - \tilde{\mu}_2)^2 = (V^T \mu_1 - V^T \mu_2)^2 \\ = V^T (\mu_1 - \mu_2) (\mu_1 - \mu_2)^T V \\ = V^T S_B V.$$

$$\text{So we have, } \tilde{s}_1^2 + \tilde{s}_2^2 = V^T S_W V \\ (\tilde{\mu}_1 - \tilde{\mu}_2)^2 = V^T S_B V$$

$$S_W = S_1 + S_2 \\ \text{(within class covariance matrix)} \\ S_B = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T \\ \text{(between class covariance matrix)}$$

Goal: $\max_V \frac{V^T S_B V}{V^T S_W V}$

This can be solved by Lagrange multipliers & eigen decomposition.

Solving this gives the following:

$$S_W^{-1} S_B V = \lambda V$$

$$\max_V V^T S_B V \text{ such that } V^T S_W V = 1$$

So the optimization problem becomes:

$$\max_V V^T S_B V = \max_V V^T \lambda S_W V \\ = \max \lambda (V^T S_W V) = \max \lambda$$

So λ is an eigenvalue of $S_W^{-1} S_B$.

Hence we should project the data points on a direction which is an eigenvector corresponding to the maximum eigenvalue of $S_W^{-1} S_B$.